La Scuola d'Italia / Dr. Huson / IB Math 11
28 October 2026 Markscheme

2.6 Quiz: Lines, Exponents and Radicals — Markscheme

Version (a)

Total: 35 marks.

1. [Maximum mark: 10]

(a) $m_{AB} = \dfrac{5-1}{4-(-4)}$ **M1**

$m_{AB} = \dfrac{4}{8} = \dfrac{1}{2}$ **A1**

A bare answer earns M0A1. Wrong method: $\frac{\Delta x}{\Delta y} = 2$: M0A0.

(b) $y - 1 = \frac{1}{2}(x+4)$ **M1**

$y = \frac{1}{2}x + 3$ **A1**

M1 for substituting their gradient and either point. The A1 needs the form $y = mx + c$. FT from (a).

(c) $m_L = -2$ **A1**

$y - 6 = -2(x-1)$ **M1**

$y = -2x + 8$ **A1**

A1 for the negative reciprocal of their gradient. Wrong methods: the perpendicular bisector of AB, $y = -2x + 3$, passes through the midpoint $(0,\ 3)$, not through C: A1M0A0; $m_L = 2$ (no sign change) gives $y = 2x + 4$: A0M1A0. FT from (a).

(d) $-2x + 8 = \frac{1}{2}x + 3$ **M1**

$\frac{5}{2}x = 5$

$x = 2, \quad y = 4 \qquad (2,\ 4)$ **A1A1**

M1 for solving their two lines together by any method (equating, substitution, GDC). A1 for each coordinate. Wrong method: the midpoint of AB, $(0,\ 3)$: M0A0A0. FT from (b) and (c).

2. [Maximum mark: 7]

(a) $2x + 3y = 47$ **A1**

$x + 4y = 41$ **A1**

(b) $x = 41 - 4y, \quad 2(41 - 4y) + 3y = 47$ **(M1)**

$5y = 35$

adult ticket $x = 13$ euros **A1**

child ticket $y = 7$ euros **A1**

M1 for any valid method, including the GDC simultaneous-equation solver; a correct GDC answer with no other working earns full marks. FT from (a).

(c) $2(13) + 7k = 61$ **M1**

$7k = 35, \quad k = 5$ children **A1**

Wrong method: swapping the prices, $2(7) + 13k = 61$*, gives* $k = 3.62$*, not a whole number: M0A0. FT from (b).*

3. [Maximum mark: 10]

(a) $(2a^3b)^2 = 4a^6b^2$ **M1**

$4a^6b^2 \times 3a^{-2}b^{-4} = 12a^4b^{-2}$ **A1**

$= \dfrac{12a^4}{b^2}$ **A1**

M1 for raising every factor of the bracket to the power 2*. The final A1 needs positive exponents only. Wrong method: leaving the* 2 *unsquared gives* $\frac{6a^4}{b^2}$*: M0A0A0.*

(b) $\left(\dfrac{8}{27}\right)^{-\frac{2}{3}} = \left(\dfrac{27}{8}\right)^{\frac{2}{3}} = \left(\dfrac{3}{2}\right)^2$ **M1**

$= \dfrac{9}{4}$ **A1**

Accept $2\frac{1}{4}$*.* 2.25 *from the GDC with no working earns M0A1. Wrong method: ignoring the negative sign gives* $\frac{4}{9}$*: M0A0.*

(c) $(3^2)^{x-1} = (3^3)^{x-3}$ **M1**

$2x - 2 = 3x - 9$ **A1**

$x = 7$ **A1**

M1 for writing both sides as powers of 3*. Wrong method:* $2x - 2 = 3x - 3$ *(the* 3 *not multiplied through) gives* $x = 1$*: M1A0A0.*

(d) $2^{x+3} = 2^{-5}$ **M1**

$x + 3 = -5, \quad x = -8$ **A1**

M1 for writing $\frac{1}{32}$ *as* 2^{-5}*. Wrong method:* $\frac{1}{32} = 2^5$ *gives* $x = 2$*: M0A0.*

4. [Maximum mark: 8]

(a) $5\sqrt{2} + 2\sqrt{2} - 4\sqrt{2}$ **M1**

$= 3\sqrt{2}$ **A1**

M1 for writing at least two of the surds as multiples of $\sqrt{2}$*. Wrong method:* $\sqrt{50 + 8 - 32} = \sqrt{26}$*: M0A0.*

(b) $9 - 3\sqrt{5} + 3\sqrt{5} - 5$ **M1**

$= 4$ **A1**

Accept $3^2 - \left(\sqrt{5}\right)^2$ *for the M1. Wrong method:* $9 + 5 = 14$*: M0A0.*

(c) $\dfrac{8}{3 - \sqrt{5}} \times \dfrac{3 + \sqrt{5}}{3 + \sqrt{5}} = \dfrac{8\left(3 + \sqrt{5}\right)}{4}$ **M1**

$= 6 + 2\sqrt{5}$ **A1**

Accept $a = 6$*,* $b = 2$*. FT from (b).*

(d) $4 = 3x\sqrt{2} - x\sqrt{2} = 2x\sqrt{2}$ **M1**

$x = \dfrac{4}{2\sqrt{2}} = \dfrac{2}{\sqrt{2}} = \sqrt{2}$ **A1**

M1 for collecting the x*-terms on one side. The A1 needs* $\sqrt{2}$*:* $\frac{2}{\sqrt{2}}$ *or* 1.41 *earns A0.*